Item/Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
			Unlikely(1)-Likely(5)	Low(1)-High(5)	Easy(1)-Hard(5)	Min(1)-Max(125)	
Body Tube:	Body tube cracks	Could prevent reuse or reduce stability	2	3	2	12	Handling the rocket carefully and controlling descent velocity
	Body tube get burnt through	Cause a hole leading to decreased stability and prevent reuse	2	3	1	6	Containing motor burn or carefully positioning motor to avoid burning
	Body tube disconnects early	Invalid competition entry as altitude will not be reached	1	5	1	5	Ensure proper connection
	Body tube doesn't separate	Safety violation as rocket will have no decreased velocity	1	5	1	5	Ensure proper connection
	Body tube fails to bond with centering rings	Motor tube disconnects leading to safety hazard	1	5	3	15	Ensure proper connection & build up epoxy
	Body tube fails to properly bond with nose cone	Rocket becomes unstable and crashes	1	5	3	15	Ensure proper connection & build up epoxy
	Rail pins create crack propagation	Could prevent reuse depending on crack size	3	3	2	18	Carfully screw into rocket
	Rail pins tear through body	Tear could prevent reuse	1	3	1	3	Ensure proper connection and screw is not fretting
	Body tube buckles	Instability; rocket crashes	1	5	1	5	Carefully handle rocket and avoid coming in contact with liquid
Nose Cone:	Nose cone disconnects during flight	Instability; rocket crashes	1	5	1	5	Ensure proper connection
	Nose cone buckles during flight	Instability; rocket crashes	1	5	2	10	Ensure material is stable & has a strong bond
	Nose cone cracks	Could prevent reuse or reduce stability	2	3	2	12	Handling the rocket carefully and controlling descent velocity
	Nose cone dents	Could prevent reuse or reduce stability	3	3	1	9	Handling the rocket carefully and controlling descent velocity
Fins:	Fins crack before/during ignition	Could lead to additional cracking upon landing; prevent reuse	4	2	2	16	Handling the rocket carefully and controlling descent velocity
	Fins chip before/during ignition	Could affect flight performance/stability	2	4	1	8	Handling the rocket carefully and controlling descent velocity
	Fins break off before/during ignition	Extreme hazard/safety concern	1	5	1	5	Handling the rocket carefully and controlling descent velocity
	Fins break off during landing	Prevents reuse entirely; defies competition rules	2	5	1	10	Handling the rocket carefully and controlling descent velocity
	Fins crack during landing	Compromises fin integrity	3	1	2	6	Handling the rocket carefully and controlling descent velocity
	Fins chip during landing	Potentially prevents reuse depending on size	4	4	1	16	Handling the rocket carefully and controlling descent velocity
	Fins fail to bond to the body tube	Fins fall off, leading to instability	1	5	1	5	Ensure proper connection & build up epoxy
	Fin material shears	Compromises fin integrity	1	2	2	4	Delicate Manufacturing & material handling
	Fins burn during ignition	Compromises fin integrity	3	3	1	9	Containing motor burn or carefully positioning fins to avoid burning
	Fins misaligned for launch	Could lead to instability	2	5	3	30	Ensure proper alignment, connection, and epoxy build-up
	Uneven fin surfaces	Compromises fin integrity and air flow	1	2	3	6	Delicate Manufacturing & material handling
	Fins flutter during flight	Causes instability	2	5	2	20	Ensure proper alignment, connection, and epoxy build-up
	Fins bend during flight	Causes instability	2	5	3	30	Ensure material is stable & has a strong bond
Bulkheads:	Bulkhead fails to bond with body tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Bulkhead fails to bond with coupler	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Bulkhead fails to bond with payload bay(s)	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Bulkhead cracks	Compromises bulkhead integrity	2	1	2	4	Handling the rocket carefully and controlling descent velocity
	Bulkhead chips	Compromises bulkhead integrity	2	4	1	8	Handling the rocket carefully and controlling descent velocity
	Bulkhead disconnects/bursts	Rocket integrity compromised causing launch failure	2	5	1	10	Ensure proper alignment, connection, and epoxy build-up
	Fretting observed between eye bolt & bulkhead	Possible disconnect of eye bolt and therefore recovery system failure	2	5	2	20	Delicate Manufacturing & material handling
	Eye bolt tears out of bulkhead	Recovery system failure	2	5	1	10	Ensure material is stable & has a strong bond
	Bulkhead burns	Compromises bulkhead integrity	1	2	1	2	Containing motor burn/boom pool
Centering Rings:	Centering ring(s) fail to bond with body tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Centering ring(s) fail to bond with motor tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Centering ring(s) crack	Compromises centering ring integrity	2	1	2	4	Handling the rocket carefully and controlling descent velocity
	Centering ring(s) chip	Compromises centering ring integrity	2	4	1	8	Handling the rocket carefully and controlling descent velocity
	Centering ring(s) disconnect/burst	Rocket integrity compromised causing launch failure	2	5	1	10	Handling the rocket carefully and controlling descent velocity
	Fretting observed between eye bolt & centering ring	Possible disconnect of eye bolt and therefore recovery system failure	2	5	2	20	Ensure proper alignment, connection, and epoxy build-up
	Eye bolt tears out of Centering ring	Recovery system failure	2	5	1	10	Ensure material is stable & has a strong bond
	Centering ring(s) burn	Compromises centering ring integrity	1	2	1	2	Containing motor burn/boom pool
Motor Retainer:	Female side of retainer disconnects from bottom centering ring	Lack of motor security; Possible motor loss/instability	2	5	1	10	Ensure proper connection & build up epoxy
	Male side of retainer unscrews during launch	Lack of motor security; Possible motor loss/instability	1	5	1	5	Ensure proper connection
	Retainer cracks	Integrity compormised	2	2	2	8	Handling the rocket carefully and controlling descent velocity
	Retainer chips	Integrity compromised	1	3	1	3	Handling the rocket carefully and controlling descent velocity
	Retainer burns and/or melts	Shape failure and possibly unable to hold motor	2	4	1	8	Containing motor burn
	Retainer bursts	Rocket integrity compromised causing launch failure	2	5	1	10	Handling the rocket carefully and controlling descent velocity
Motor:	Motor explodes	Invalid competition entry	1	5	1	5	N/A
	Ejection charge fails	Drogue does not deploy	2	4	1	8	Backup ejection charge for drogue
	Motor case fails	Rocket may explode	1	4	5	20	N/A
	Motor fails to ignite	Rocket does not launch	1	5	1	5	Ensure igniter isn't damaged
	Motor falls out	Rocket loses propulsion	1	5	5	25	Ensure retaining spring and centering rings are undamaged

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			Unlikely(1)-Likely(5)	Low(1)-High(5)	Easy(1)-Hard(5)	Min(1)-Max(125)	
	Motor wall burn-through	Heat may burn through rocket tube	1	3	5	15	N/A
	Nozzle blow-out	Motor loses thrust	1	5	5	25	N/A
	Premature ejection	Target altitude not reached	1	5	5	25	N/A
Shock Chord:	Shock cord snaps	Parachute loss	2	5	1	10	Ensure shock cord isn't damaged before launch
	Zippering	Body tube damage	3	1	1	3	Ensure zipper protectors are placed correctly
	Shock cord gets tangled	Shock cord doesn't reach full length	2	2	3	12	Ensure shock cord is folded correctly
	Shock cord gets burned	Possible parachute loss	1	4	3	12	Ensure wad is low over motor
Parachute:	Parachute tears	Rocket can't fully slow down	3	5	4	60	Ensure parachute is intact before launch
	Parachute doesn't fully deploy	Rocket can't fully slow down	3	5	4	60	Ensure parachute lines aren't tangled before launch
	Burned parachute	Rocket can't fully slow down	2	3	5	30	Ensure wad fully covers parachute
	Parachute lines tangled	Rocket can't fully slow down	2	3	3	18	Ensure parachute lines aren't tangled before launch
	Improper packing	Parachute may not deploy	1	3	1	3	Ensure parachute is packed correctly
	Improper folding	Parachute may not deploy correct	1	3	1	3	Ensure parachute is folded correctly
	Wad plugging	Parachute may not deploy	1	5	2	10	Ensure wad can move freely in body tube
	Broken parachute lines	Possible parachute loss	2	4	1	8	Ensure parachute lines are intact before launch
	Inadequate parachute size	Rocket can't fully slow down	2	5	4	40	Test with simulators to attain correct parachute size
	Early deployment	Rocket drifts too far	4	5	5	100	Ensure altimiter is set correctly
	Late deployment	Rocket can't fully slow down	4	5	5	100	Ensure altimiter is set correctly
Water Vessel:	Vessel Structural Damage	Leakage resulting in disqualification	1	5	5	25	Ensure vessel can withstand forces with margin
	Vessel Bursting	Leakage resulting in disqualification	1	5	5	25	Ensure vessel can withstand forces with margin
	Vessel Thermal Damage	Structural integrity of the vessel compromised	1	3	4	12	Ensure vessel is shielded from direct thermal loads
	Vessel Unplanned Detachment	Loose vessel which could become a projectile	2	4	5	40	Ensure vessel is properly seated in place
	Caps Improperly Sealed	Leakage resulting in disqualification	2	5	3	30	Properly inspect sealing prior to launch
	Caps Loosening due to Vibration	Leakage resulting in disqualification	2	5	5	50	Ensure caps are carefully and properly installed
	Caps damaged due to Over-tightening	Leakage resulting in disqualification	2	5	4	40	Ensure caps are carefully and properly installed
	Supports Structural Damage	Loose vessel which could become a projectile	1	4	5	20	Ensure supports can withstand forces with margin
	Supports Unplanned Detachment	Loose vessel which could become a projectile	1	4	5	20	Ensure supports can withstand forces with margin
	Supports Improperly Manufactured	Improper fittings exacerbating other issues	3	3	1	9	Ensure the supports are manufactured within specified margins
						0	
						0	
Egg Vessel:	Failure to protect the eggs						
	Failure to dampen vibrations	Can cause cracks	3	5	4	60	Testing to check dampening is effictient
	Failure to dampen impact forces	Can cause cracks	3	5	4	60	Testing to check dampening is effictient
	Faliure to properly secure eggs	Can cause cracks	2	5	5	50	Test if egg pucks will rotate in capsule or using two rods in capsule
	Failure to unload under five minutes						
	Capsule stuck in rocket	Can't load/ unload under 5 minutes	3	5	1	15	Test how capsule fits in rocket and practice loading and unloading
	Capsule not opening	Can't load/ unload under 5 minutes	1	5	1	5	Repeated practice of opening and closing the capsule
	Egg pucks take too long to stack	Can't load under 5 minutes	1	5	1	5	Practice stacking of pucks
	Egg pucks take too long to unstack	Can't unload under 5 minutes	1	5	1	5	Practice unstacking of pucks
Avionics Bay:	Avionics bay integrity failure on ground hit	High potential for electronics damage	2	4	2	16	Minimize ground-hit velocity
	Avionics bay integrity failure on recovery deployment	High potential for electronics damage	2	4	2	16	Minimize ground-hit velocity
	U-Bolt shear out	Recovery failure; potential freefall	3	5	3	45	Use thick bulkheads and separate at apogee
	E-sled improperly secured	Probable component deseating or additional vibration stress	2	3	3	18	Ensure all parts are within proper tolerance and torque
	Components improperly secured to e-sled	Probable component deseating or additional vibration stress	2	3	3	18	Ensure all parts are within proper tolerance and torque
Avionics Electronics:	Premature battery depletion	Complete loss of FC and camera functions	2	5	3	30	Size battery (mAh rating) for long pad operations
	Flight controller dropout (malfunction)	Loss of reliable flight event control and data logging	1	5	2	10	Ensure proper handling and operation procedure is folloowed
	Screw terminal wire de-seating	Loss of power to electrical functions mid-flight	3	5	2	30	Secure all wires with items like zipties
	Camera dropout (malfunction)	Loss of recorded video (bonus points)	2	2	2	8	Ensure large enough SSD storage and operation
	Camera fogging obstruction	Loss of video quality	1	1	4	4	Minimize temperature change and visually inspect
	Electrical component condensation failure (Humidity)	Potential loss of reliable flight event control and data logging	1	5	4	20	Minimize temperature change to prevent condensation
	Electrical component overheat failure (Temperature)	Potential loss of reliable flight event control and data logging	1	5	4	20	Ensure proper component venting and spacing
	Black powder ignition failure	Failure of primary recovery deployment	2	5	4	40	Keep powder dry and securely packed
	Audio recovery device failure	Loss of ground recovery aide	3	2	2	12	Only enable device as close to before launch as possible

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	Unintentional arming switch toggle	Potential premature recovery event	1	4	1	4	Ensure arming switch is secured in either operating position